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| **Fiscal Year:** | FY 2013 | **Task Last Updated:** | FY 09/23/2013 |
| **PI Name:** | Fernandez-Pello, Carlos Ph.D. | | |
| **Project Title:** | Material Ignition and Suppression in Space Exploration Atmospheres | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | COMBUSTION SCIENCE | | |
| **Program/Discipline--Element/Subdiscipline:** | COMBUSTION SCIENCE--Combustion science | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | ferpello@me.berkeley.edu | **Fax:** | FY 510 642-6163 |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 510 642-6554 |
| **Organization Name:** | University of California, Berkeley | | |
| **PI Address 1:** | Department of Mechanical Engineering | | |
| **PI Address 2:** | 6105A Etcheverry Hall | | |
| **PI Web Page:** | | | |
| **City:** | Berkeley | **State:** | CA |
| **Zip Code:** | 94720-1740 | **Congressional District:** | 9 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2009 Combustion Science NNH09ZTT001N |
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| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 2 | **Monitoring Center:** | NASA GRC |
| **Contact Monitor:** | Urban, David | **Contact Phone:** | 216-433-2835 |
| **Contact Email:** | david.l.urban@nasa.gov | | |
| **Flight Program:** | | | |
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| **Key Personnel Changes/Previous PI:** | | | |
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| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | Central Objectives: The next generation of NASA spacecraft will use lower pressures and increased oxygen concentrations than current designs. Although there is insufficient knowledge about fire behavior of materials in these Space Exploration Atmospheres (SEA), recent research suggests that material ignitability is increased in SEA. These results imply that next generation spacecraft present a greater fire safety hazard than current designs. Although the most effective fire safety strategy is to prevent ignition, if ignition does occur, it is critical to rapidly extinguish the fire. Therefore, an experimental investigation into ease of ignition and fire suppression characteristics of materials under SEA is proposed.<br>Methods/techniques: The proposed experimental apparatus is simple and easily incorporated in the Combustion Integrated Rack on the International Space Station. It consists of a small-scale cylindrical flow duct that has fuel samples | | |

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| **Task Description:** | attached to its inside wall. A cylindrical radiant heater is located at the center of the duct. An oxidizer gas flows at down the length of the tube. Electrical igniters are placed downstream of each fuel sample. Time to ignition and the time between introduction of suppression agent and extinction are measured as a function of environmental conditions. The normal gravity and microgravity experimental programs will be supported by a comprehensive model of piloted ignition and extinction of materials in SEA. Microgravity experiments will be used to validate the model, which in turn will be used to predict the fire behavior of different combustible materials in SEA.<br><br>Perceived Significance: Currently, there exists no testing methodology specifically designed to determine the fire hazards of materials under those conditions. The proposed work will fill that void and provide additional information about the effect of SEA on the flammability and suppression of materials. It will also provide guidance to interpret and extend current NASA testing procedures to SEA.<br><br>[Ed. note: Continues as grant NNX13AL10A, Material Ignition and Suppression Test (MIST) in Space Exploration Atmospheres] |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The project is intended to enhance the ability to predict ignition, suppression, and spread characteristics of materials in their end use configuration. This approach is called performance based design in terrestrial fire protection engineering. Computational support for this approach requires detailed information about material thermal and combustion properties. Our work on MIST will use experimental data to validate computer models attempting to predict fire behavior using material property estimation techniques. These techniques are currently a focus of research in fire protection engineering. |
| **Task Progress:** | The Material Ignition and Suppression in Space Exploration Atmospheres project was a joint experimental and numerical investigation of ignition and suppression of radiatively heated, piloted, solid fuel samples.<br>The current experimental investigation of the piloted ignition of cylindrical samples of Poly(methyl_methacrylate) (PMMA) under an externally applied radiant flux. The PMMA rods are placed vertically in a natural convection air flow. The work expands upon previous investigations that were concerned with the piloted ignition characteristics of flat samples of PMMA or similar. The primary novel aspect of this experimental investigation is the cylindrical symmetry of the samples and the associated transverse curvature of the boundary layer that acts to effect thermo-chemical characteristics of the sample heating and ignition. There seem to be (1) slightly longer ignition delay times when compared to similar conditions for flat samples, and (2) similar fuel mass flux at ignition. Both of these phenomena can be attributed at least in part to the symmetrical configuration and a thinner boundary layer present due to the curvature of the sample surface. It is proposed that a thinner boundary layer results may result in greater convective cooling of the sample due to said thinner boundary layer, thus increasing the time to ignition, without significantly affecting the critical mass flux.<br><br>These results can potentially be attributed to the thinner boundary layer leading to more convective cooling of the sample that in turn would increase the time taken to reach the flash and then fire points of the material. More work will be required to ascertain if these longer ignition times are sustained over cylinders of different radii and lengths, as well as whether the mass flux at ignition continues to be constant.<br><br>Modeling has been carried out in conjunction with the above experimental work to simulate the MIST experiments. One version of the current model currently in use was developed to simulate a 2-D solid fuel slab using the Fire Dynamics Simulator (FDS) version 6R3 in its Direct Numerical Simulation (DNS) mode. A second considers the geometry of the current experiment using a long antisymmetric fuel rod in a duct similar to that in the current iteration of the experiment. Both versions of the model simultaneously consider the processes in the solid and in the gas phases. The solid phase decomposition is modeled with a single step global Arrhenius reaction rate. Oxidative pyrolysis is not considered and the in-depth formed pyrolyzate is assumed to flow unrestricted through the solid combustible. The gas phase kinetics are modeled with a single-step second order Arrhenius reaction rate. Although the current model may simulate well the cylindrical experiments, it may be necessary to confirm this and implement any necessary modification.<br><br>2-D Modeling results seem to indicate that buoyancy plays an important role through the Rayleigh-Taylor instability at normal gravity, particularly once the flame is established over the PMMA surface. However, the Rayleigh-Taylor instability disappears for zero gravity conditions and the flame shows a distinct round, more uniform shape. Furthermore, ignition occurs significantly earlier at 0-g than at 1-g conditions.<br><br>In comparison, the cylindrical model does not yet exhibit predictive results of the same caliber, yet both the flat slab and cylindrical rod models are moving towards becoming useful predictive tools for the study of ignition and flame spread phenomena. Reasons for the lag in the cylindrical models predictive ability may be potentially attributable to how the incident heat flux boundary condition was imposed on the surface, as well as how it does not currently account for edge effects due to the finite length of the heater. Another possible concern may be the asymmetry implied by the model, which may not be accounting for the thinner boundary layer of the rod, thus over predicting the convective losses, leading to the sample to heat faster.<br><br>[Ed. note 9/23/2013: Continues as grant NNX13AL10A, Material Ignition and Suppression Test (MIST) in Space Exploration Atmospheres; see that task with same PI for subsequent reporting] |
| **Bibliography Type:** | Description: (Last Updated: 02/15/2026) |
| **Abstracts for Journals and Proceedings** | Fereres S, Fernandez-Pello C, Urban D, Ruff G. "Identifying the Roles of Microgravity and Reduced Pressure on the Ease of Ignition of Solid Combustibles." 42nd International Conference on Environmental Systems, San Diego, CA, July 15-19, 2012.<br>42nd International Conference on Environmental Systems, San Diego, CA, July 15-19, 2012. AIAA paper 2012-3537. http://dx.doi.org/10.2514/6.2012-3537 , Jul-2012 |

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| **Articles in Peer-reviewed Journals** | Fereres S, Lautenberger C, Fernandez-Pello C, Urban DL, Ruff GA. "Understanding ambient pressure effects on piloted ignition through numerical modeling." Combustion and Flame. 2012 Dec;159(12):3544-53. http://dx.doi.org/10.1016/j.combustflame.2012.08.006 , Dec-2012 |